Aspect-Based Sentiment Analysis

SemEval-2014 Task 4 (Restaurants) · Fine-tuned transformer · Ray Tune hyperparameter search · Trained on Kaggle (2 × NVIDIA T4)

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Abstract

This project builds an aspect-based sentiment analysis (ABSA) system that decides whether a review expresses a **positive**, **negative** or **neutral** opinion toward a *specific aspect term* in a sentence, rather than toward the sentence as a whole. We cast the task as sentence-pair classification and fine-tune a pre-trained RoBERTa-base encoder on the SemEval-2014 Restaurants data. Hyperparameters were selected with a 20-trial Ray Tune search (ASHA scheduler) across two backbones, using a sentence-grouped validation split that prevents leakage. The final model, retrained on all labelled data, reaches **86.25% accuracy and 0.7883 Macro-F1** on the official gold test set, with ROC AUCs of 0.96, 0.98 and 0.90 for the positive, negative and neutral classes. Two robustness experiments (20 challenge sentences and 20 minimal-edit pairs) show that the model handles most contrastive and out-of-domain cases and keeps its prediction in 17 of 20 perturbation pairs.

1. Introduction

Overall sentiment is often not enough. In the sentence "The meal tasted great, but the waiter's attitude ruined the evening," the opinion about the *meal* is positive while the opinion about the *waiter* is negative. A useful system must therefore model the relation between a sentence and a target aspect.

The goals of the project were to:

- extract one training example per (sentence, aspect) pair from the SemEval-2014 XML data and analyse it;
- design and train a transformer-based three-class aspect polarity classifier;
- evaluate it with metrics suited to imbalanced data, with emphasis on Macro-F1;
- analyse errors and test robustness on new, challenging sentences.

2. Data

2.1 Extraction

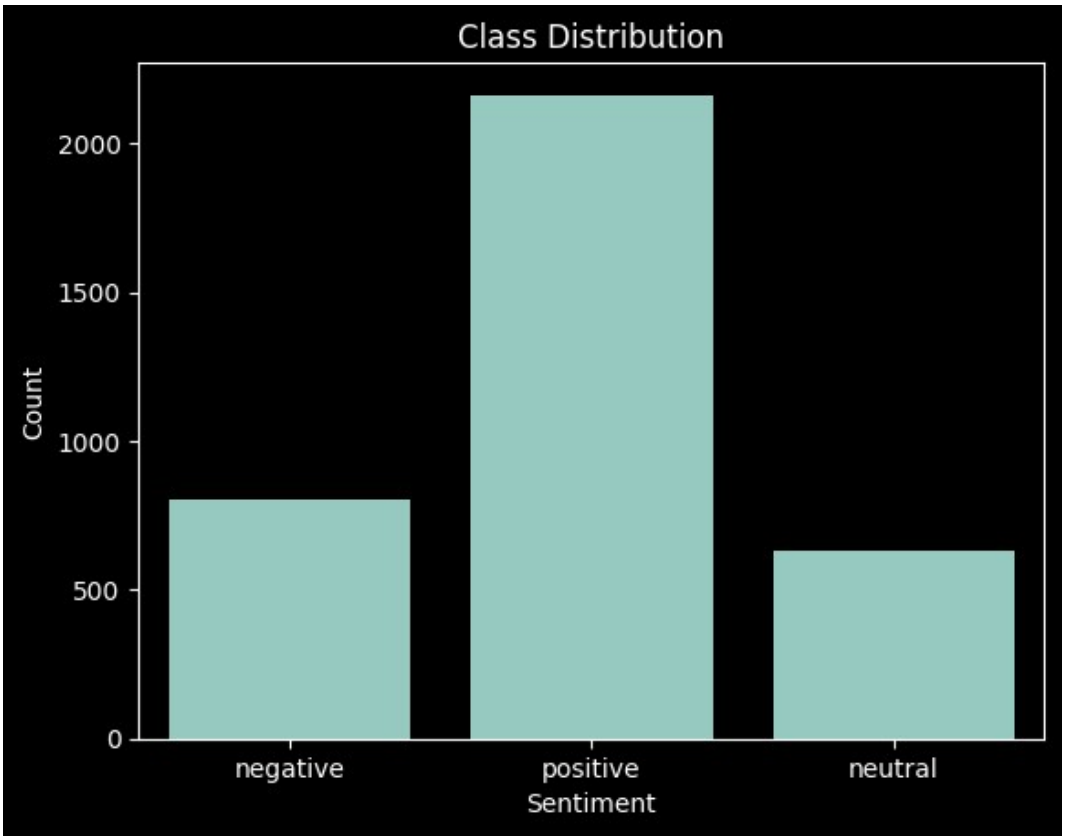
The SemEval-2014 Task 4 Restaurants XML files are parsed with `ElementTree`. Every `aspectTerm` in every sentence becomes one sample with three fields: `sentence`, `aspect` and `polarity`. Because a sentence can contain several aspects, it can produce several samples. Aspect terms labelled `conflict` are removed, and sentences with no aspect terms are skipped.

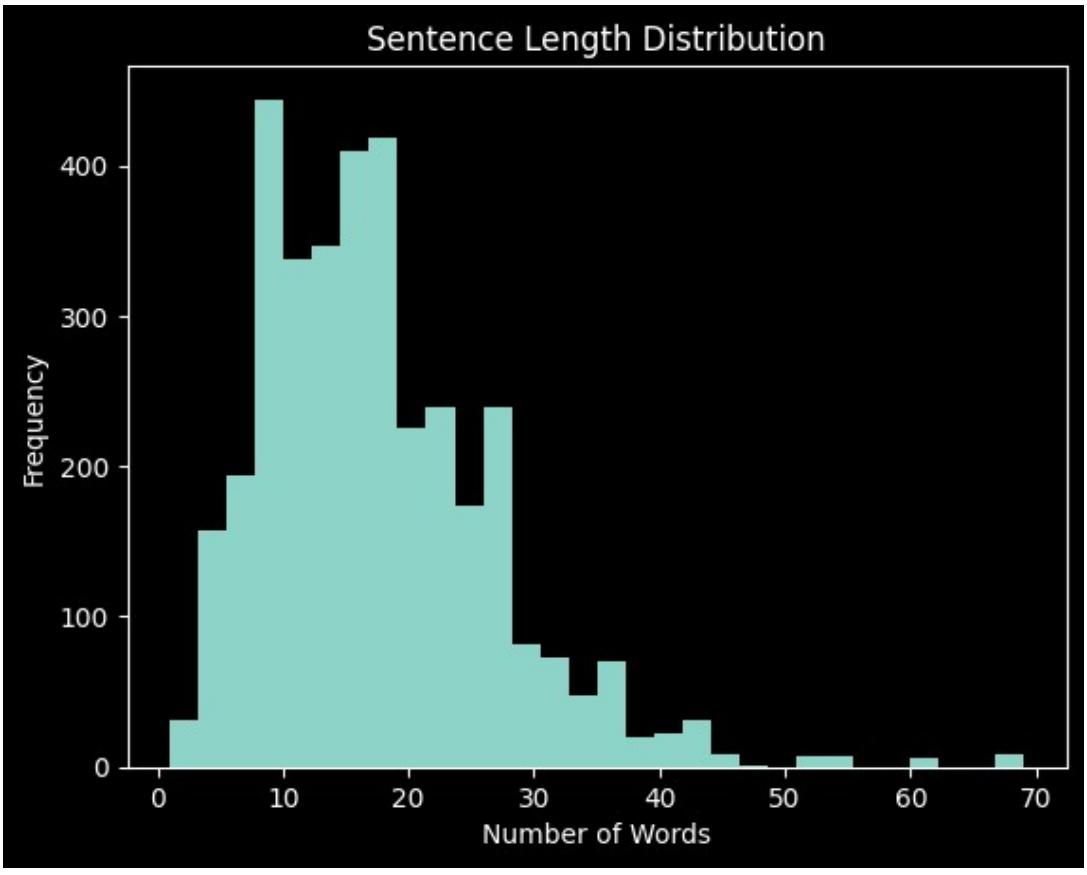
Split	Samples	Positive	Negative	Neutral
Train (Restaurants_Train_v2)	3,602	2,164 (60.1%)	805 (22.3%)	633 (17.6%)
Test (Restaurants_Test_Gold)	1,120	728 (65.0%)	196 (17.5%)	196 (17.5%)

2.2 Exploratory analysis

- **Class distribution.** Positive is the majority class in both sets, so accuracy alone would be misleading. This motivated Macro-F1 as the main metric and class weighting during training.
- **Sentence length.** Most sentences contain between 5 and 30 words, with a peak around 8 to 18 and a long tail up to about 70 words. A maximum input length of 128 tokens therefore covers essentially all samples without losing content.
- **Word clouds.** Word clouds were generated separately for positive, negative and neutral samples after lowercasing, removing non-letters and removing stopwords. They show clearly different vocabularies per polarity, such as praise words for positive samples and service and wait-time complaints for negative ones.

All figures are available in the repository (`assets/` folder and notebook).


Class Distribution
Count
2000
1500
1000
500
0
negative
positive
neutral
Sentiment


Sentence Length Distribution
Frequency
400
300
200
100
0
0
10
20
30
40
50
60
70
Number of Words

3. Method

3.1 Input representation

Sentence and aspect are encoded as a tokenizer text pair, `tokenizer(sentence, aspect)`, which for RoBERTa produces `<s> sentence </s></s> aspect </s>`. The model can therefore attend jointly to the context and the target term. Sequences are padded to 128 tokens, and labels are mapped as positive = 0, negative = 1, neutral = 2.

3.2 Model

We use a pre-trained encoder with a sequence-classification head (`AutoModelForSequenceClassification`, `num_labels=3`). The pre-trained language-model head is discarded and a new three-class classifier is initialised, which is the intended setup for fine-tuning. Training runs in FP32.

3.3 Training objective and optimisation

- **Loss:** cross-entropy with inverse-frequency class weights, `w_c = N / (3 · n_c)`, computed on the training set, to counter the dominance of the positive class.
- **Optimiser:** AdamW with tuned weight decay.
- **Schedule:** linear warm-up over the first 10% of steps followed by linear decay.
- **Stability:** gradient clipping at a maximum norm of 1.0, and automatic detection of non-finite loss or gradient values so that a diverging trial fails fast instead of reporting misleading numbers.
- **Model selection metric:** validation Macro-F1 (best epoch).

3.4 Validation split

Because several samples can share one sentence, a random split would leak the same sentence into both train and validation. We use `StratifiedGroupKFold` (5 folds, shuffled, seed 42), grouped by sentence text, and take the first fold. This gives a leakage-free split that approximately preserves class proportions.

Split	Samples	Positive	Negative	Neutral
Train	2,878	59.6%	22.3%	18.0%
Validation	724	61.9%	22.4%	15.7%

3.5 Hyperparameter search

Hyperparameters were searched with Ray Tune using the ASHA scheduler (metric: validation Macro-F1, maximised) and one GPU per trial on 2 × NVIDIA T4. 20 trials were sampled from the following space, and **all 20 completed without errors**.

Parameter	Search space
Backbone	`roberta-base`, `microsoft/deberta-v3-base`
Learning rate	log-uniform, 1e-5 to 5e-5
Batch size	16, 32
Weight decay	0.01, 0.1
Fixed	4 epochs, warm-up 10%, max length 128, gradient clip 1.0, class-weighted loss, seed 42

4. Experimental results

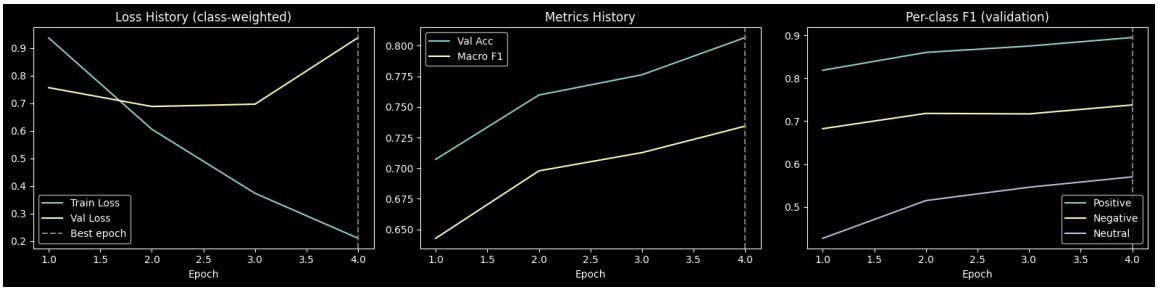
4.1 Hyperparameter search outcome

Rank	Backbone	Best validation Macro-F1	Best epoch	Learning rate	Batch size	Weight decay
1	roberta-base	**0.7343**	4	3.60e-05	16	0.01
2	roberta-base	0.7247	4	4.76e-05	32	0.1
3	roberta-base	0.7246	4	1.32e-05	16	0.1

RoBERTa-base configurations took the top three places, and the three best configurations perform similarly, which indicates that the setup is not very sensitive to the exact hyperparameters. The selected configuration is therefore a robust choice: `roberta-base`, learning rate 3.60e-05, batch size 16, weight decay 0.01, 4 epochs. (The full table of all trials is stored in `all_models_results.csv`.)

4.2 Training curves

For the selected trial, training loss decreases steadily from about 0.94 to about 0.21 over four epochs. Validation accuracy rises from about 0.71 to about 0.81, validation Macro-F1 from about 0.64 to 0.73, and every per-class validation F1 improves in every epoch (at epoch 4, roughly 0.89 for positive, 0.74 for negative and 0.57 for neutral). Validation loss is lowest around epochs 2 and 3 and rises in the last epoch while the classification metrics keep improving, a common pattern when a model becomes more confident on its mistakes.



4.3 Final model

The selected configuration was retrained on all 3,602 labelled samples (train plus validation) for four epochs, using the same settings.

Epoch	Training loss	Training accuracy
1	0.9057	0.6338
2	0.6068	0.7965
3	0.3786	0.8853
4	0.2265	0.9389

4.4 Test-set performance

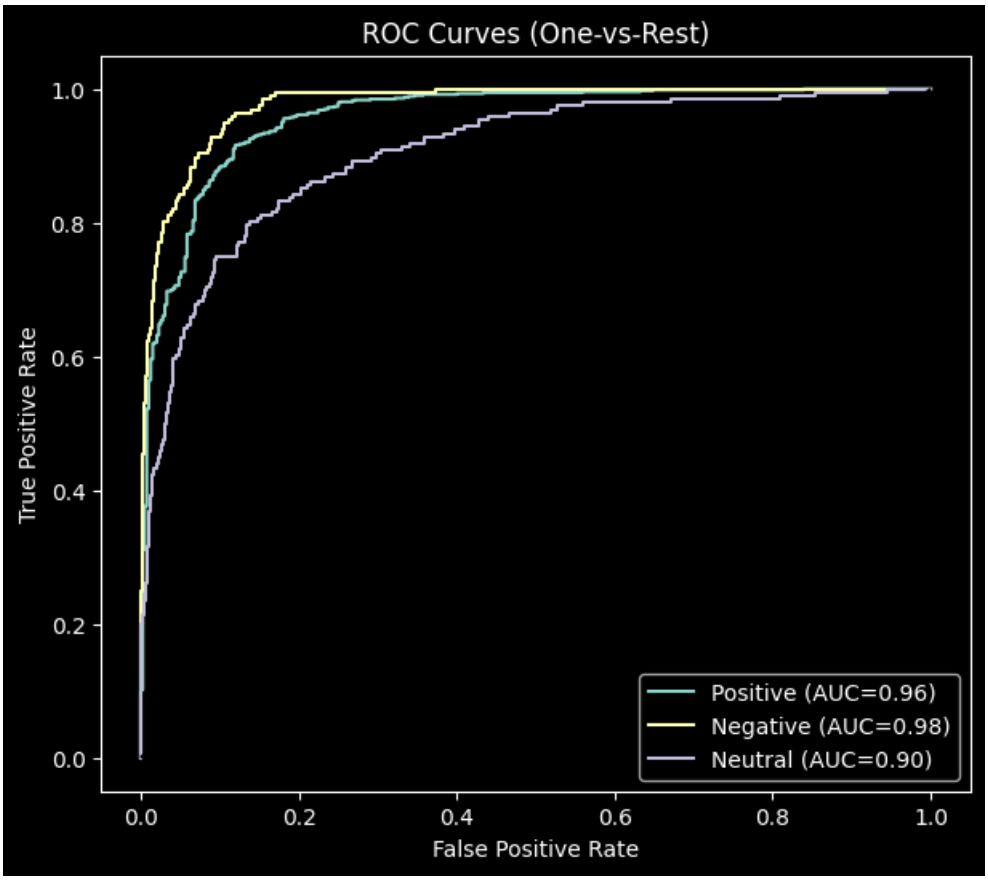
The final model was evaluated once on the official SemEval-2014 gold test set (1,120 samples).

Class	Precision	Recall	F1	Support
Positive	0.8986	0.9615	0.9290	728
Negative	0.7970	0.8214	0.8090	196
Neutral	0.7554	0.5357	0.6269	196
Macro average	**0.8170**	**0.7729**	**0.7883**	1,120

Class	Precision	Recall	F1	Support
Weighted average	0.8558	0.8625	0.8551	1,120

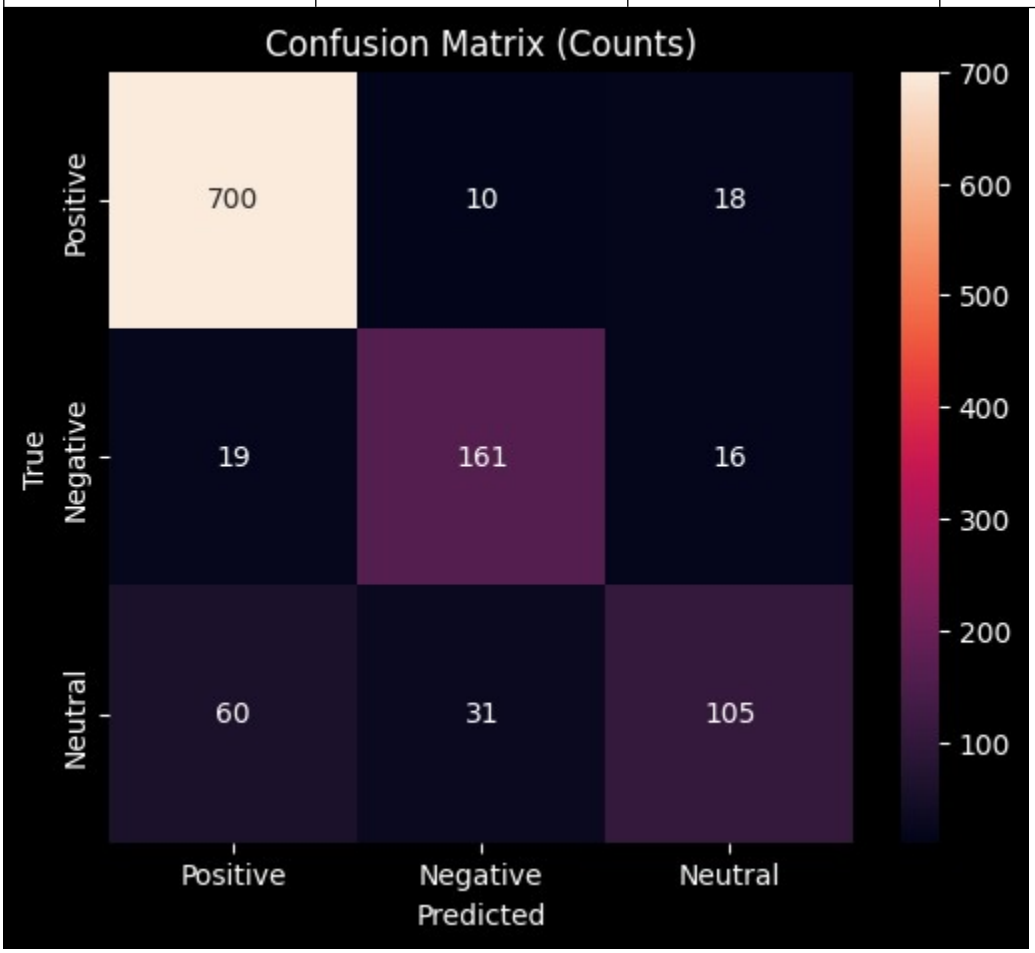
Global metric	Value
Accuracy	0.8625
Macro-F1	0.7883
Weighted F1	0.8551
Cohen's kappa	0.7220
Matthews correlation coefficient	0.7263
ROC AUC (one-vs-rest)	Positive 0.96, Negative 0.98, Neutral 0.90

A kappa and MCC above 0.72 indicate substantial agreement with the gold labels well beyond chance, and the high AUCs show that the model ranks the correct class well even where its hard decision is uncertain.



Confusion matrix (rows = true class, columns = predicted class):

True \ Predicted	Positive	Negative	Neutral
Positive	700	10	18
Negative	19	161	16
Neutral	60	31	105



Positive and negative aspects are identified accurately: 96% of positive and 82% of negative samples are recovered, and confusion between the two opposite polarities is very small (10 and 19 samples). The remaining errors are concentrated in the neutral class, where 60 samples are predicted as positive and 31 as negative. Neutral precision is 0.76, so a neutral prediction is usually right; the model is simply conservative about assigning it.

5. Error analysis

Five misclassified test samples were examined in detail.

#	Sentence	Aspect	True	Predicted	Probable cause
1	Certainly not the best sushi in New York, however, it is always fresh, and the place is very clean, sterile.	place	Positive	Negative	Negation ("not the best") and the word "sterile" suggest negativity; the contrast introduced by "however" and the praise "clean" are outweighed.
2	While there's a decent menu, it shouldn't take ten minutes to get your drinks and 45 for a dessert pizza.	dessert pizza	Neutral	Positive	The complaint concerns waiting time, not the pizza itself. The nearby word "decent" is associated with the aspect, and the model cannot separate which opinion targets which term.
3	How pretentious and inappropriate for MJ Grill to claim that it provides power lunch and dinners!	lunch	Negative	Positive	Criticism of a *claim*; the surface wording around the aspect ("power lunch") looks positive or neutral.
4	(same sentence)	dinners	Negative	Positive	Same cause as above.
5	Entrees include classics like lasagna, fettuccine Alfredo and chicken parmigiana.	Entrees	Neutral	Positive	Purely descriptive sentence with no opinion words; "classics" has a positive connotation and the training data has a positive majority.

Common themes. The remaining errors involve phenomena that are hard even for strong models: opinions that target a different phrase than the aspect, irony and implicit criticism, nested contrast and negation, and descriptive text without explicit sentiment. This is consistent with the confusion matrix, where the largest error cell is neutral predicted as positive.

6. Robustness evaluation

6.1 Twenty challenge sentences

Twenty new sentences were written to stress the model with contrast, conditionals, double negation, implicit sentiment and aspects outside the usual restaurant vocabulary (for example app, AC, delivery, seal). Eight representative cases are shown below; the full set of 20 is in the notebook. Model predictions:

Aspect	Sentence	Prediction
waiter	The meal tasted great, but the waiter's attitude ruined the evening.	Negative
dessert	I loved the main course; dessert, however, tasted like it came straight from a freezer.	Negative
food	I thought the food was good, but I'd hesitate to call it healthy.	Positive
chef	The instructions for dietary requests were confusing, yet the chef managed to accommodate us.	Positive
delivery	They claim 24/7 delivery; I called at 2 AM and no one picked up.	Negative
AC	Handles rush hour fine, but the AC fails when the crowd grows.	Negative
specials	Prices are reasonable, unless you order the specials, then it gets expensive.	Neutral
app	The app launches instantly, except when it forces an update and stalls.	Positive

Observations. The model correctly handles most "positive but negative" contrast sentences, correctly assigns the clause that actually mentions the aspect (for example food versus waiter in one sentence), and generalises to aspects far from the training vocabulary such as AC, app, seal and delivery. It also uses context well in cases like "chef" (positive despite confusing instructions) and "food" (positive despite the health remark). The most demanding items are conditional or exception clauses ("unless you order the specials") and sentences whose opinion about the aspect is genuinely mixed (ambience, menu photos, app), where even human annotators could disagree.

6.2 Minimal-edit adversarial pairs

Twenty pairs of sentences were created in which sentence B is a small edit of sentence A that is not intended to change the opinion toward the aspect. The model's prediction was **unchanged in 17 of 20 pairs** (85%), covering aspects such as sauce, cocktails, waiter, pizza, ambiance, sushi, prices, staff, delivery, salad, music, burger, dessert, table, soup, service and view. It changed in three pairs:

Aspect	Prediction on A	Prediction on B
coffee	Positive	Negative
hostess	Negative	Positive
fries	Positive	Negative

These three cases mark the limits of the model's robustness to surface-level wording changes and are a useful starting point for adversarial data augmentation.

7. Conclusion

The project delivers a complete aspect-based sentiment pipeline: a RoBERTa-base sentence-pair classifier that reaches **86.25% accuracy and 0.7883 Macro-F1** on the official gold test set. It separates positive from negative aspects very well (only 10 and 19 cross-confusions out of 924 samples) and ranks classes reliably (AUC 0.96, 0.98 and 0.90). The main reasons are a sentence-grouped validation split, a 20-trial search over two backbones, stable class-weighted training, and an evaluation that goes beyond one score to include error analysis and two robustness tests.

The clearest opportunity is the neutral class (F1 0.63, recall 0.54), because neutral mentions often contain no opinion words and are easily read as positive. Natural next steps are a larger neutral class weight or focal loss, marking the aspect span with special tokens, multi-seed tuning with an ensemble of the best configurations, adversarial augmentation based on the pairs that flipped, and evaluation on the Laptops domain.

Everything was run on a Kaggle notebook with 2 × NVIDIA T4 GPUs and seed 42. The notebook, `all_models_results.csv`, `best_config.json` and the saved model (`best_final_model`) reproduce the reported numbers from the SemEval-2014 Restaurants training and gold test files.